Section A

Concept Application

SCENARIO 1

Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Question:

What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility

1. **CPU:** Choose a fast, multi-core processor like Intel Core i5 or AMD Ryzen 5 to run multiple applications smoothly.

2. **RAM:** At least 16 GB RAM is recommended so programming software, browsers, and video calls can run together without slowing down.

3. **Storage:** Choose an SSD instead of an HDD because it starts the computer and opens applications faster. A 512 GB SSD is a good choice.

4. **Compatibility:** Make sure the CPU, RAM, and storage are supported by the motherboard and are compatible with each other.

SCENARIO 2

After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Question:

Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

1. **CPU**: Antivirus scans files and programs in the background, using CPU power and making other applications slower.

2. **RAM:** Antivirus uses RAM to run its services. This leaves less memory available for other programs.

3. **Storage:** Antivirus scans files on the drive, using storage read/write resources. This can make opening files and applications slower, especially during full scans.

SCENARIO 3

A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question:

Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

1. **Check connections:** Unplug the keyboard and printer, then connect them again to different USB ports.
2. **Check Device Manager:** Open Device Manager and look for devices with a yellow warning sign.
3. **Update drivers:** Right-click the device and select Update driver. Install the correct driver from the manufacturer's website if needed.
4. **Check Windows updates:** Install pending Windows updates that may fix compatibility issues.
5. **Restart the computer:** Restart the system and check whether both devices work properly.
6. **Test again:** Connect the devices and print a test page or check keyboard typing to confirm the problem is fixed.

SCENARIO 4

Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question:

Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

1. **UPS (Uninterruptible Power Supply):** Provides temporary battery power during a power cut, giving employees time to save their work and shut down computers safely.

2. **Surge Protector:** Protects computer hardware from sudden voltage spikes that can damage components like the motherboard and power supply.

SCENARIO 5

New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they

connect to the network.

Question:

Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

1. **DNS (Domain Name System):** DNS converts server names into IP addresses. If employees can access the server by IP but not by name, the DNS service may be misconfigured.

2. **DHCP (Dynamic Host Configuration Protocol):** DHCP automatically assigns IP addresses and other network settings to devices. If devices fail to obtain a valid IP address, DHCP may not be working correctly.

3. **How they work together:** DHCP provides devices with IP addresses and DNS server details, while DNS helps them find internal systems by name.

Conclusion:

DNS is likely misconfigured because server names are not resolving. DHCP may also have a problem because some devices cannot obtain valid IP addresses.

SCENARIO 6

After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Question:

Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

1. **Check connectivity:** Ping the internal application server to see whether it is reachable.

2. **Check firewall rules:** Review the firewall rules to see if they are blocking traffic to the application server.

3. **Check ports:** Verify that the required ports, such as HTTP (80) or HTTPS (443), are allowed.

4. **Check firewall logs:** Look for blocked connections from employees' computers to the server.

5. **Fix the rule:** Add or correct a firewall rule to allow legitimate traffic to the application server.

6. **Test again:** Try accessing the application server from an employee's computer to confirm that it works.

Section B

Practical Application

Task 1: Workstation Assembly



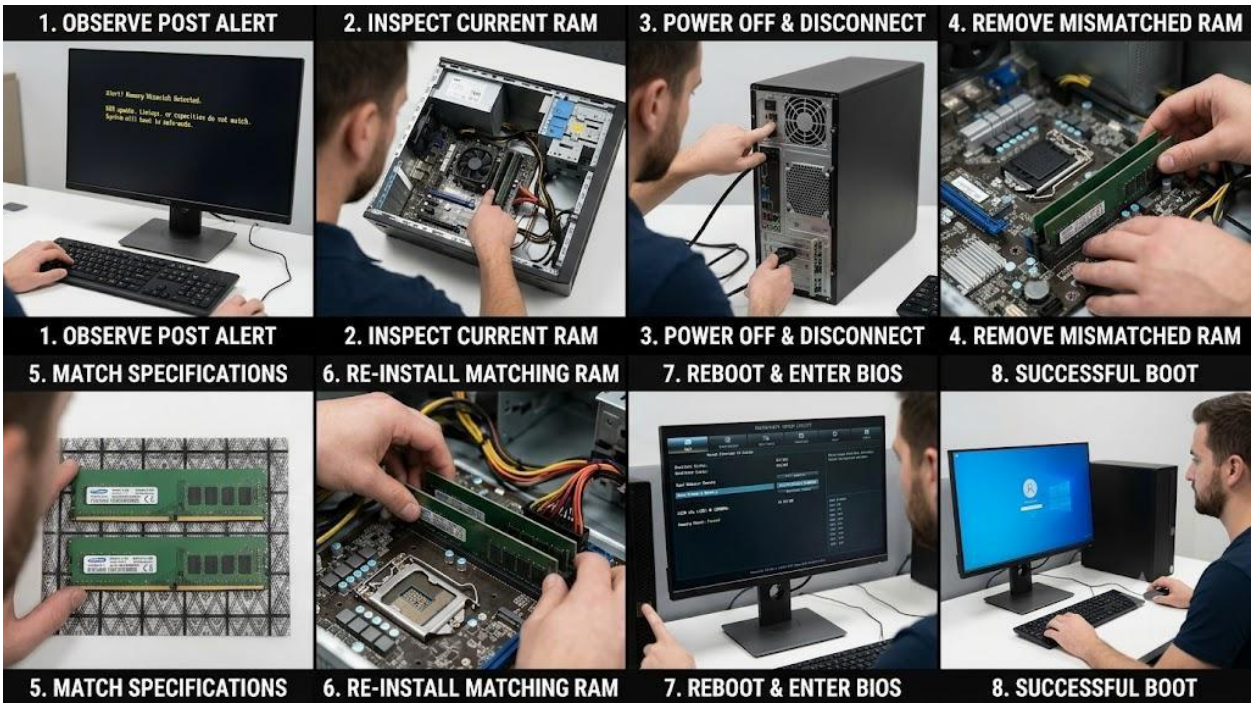
1. **Pre-assembly (Out of Case):** Placed the motherboard on a flat surface, installed the CPU, applied thermal paste, attached the cooler/fan, and clicked the RAM into place.

2. **Mounting Hardware:** Prepped the chassis, secured the populated motherboard onto its standoffs, mounted the power supply unit (PSU), and fastened the storage drives.

3. **Expansion & Cabling:** Installed the GPU into the PCIe slot, connected all power lines (24-pin, CPU, GPU), front-panel pins, and storage data cables, then tied back excess wires for clean airflow.

4. **POST Verification:** Plugged in the display and power cables, powered on the system, and confirmed it successfully completed POST by booting into the BIOS screen.

Task 2: RAM Mismatch Troubleshooting



1. **Identify Error:** Booting triggered a BIOS POST alert indicating mismatched memory speed, capacity, or timings.

2. **Isolate Hardware:** Shut down the workstation, disconnected power, and removed the mismatched RAM modules from the slots.

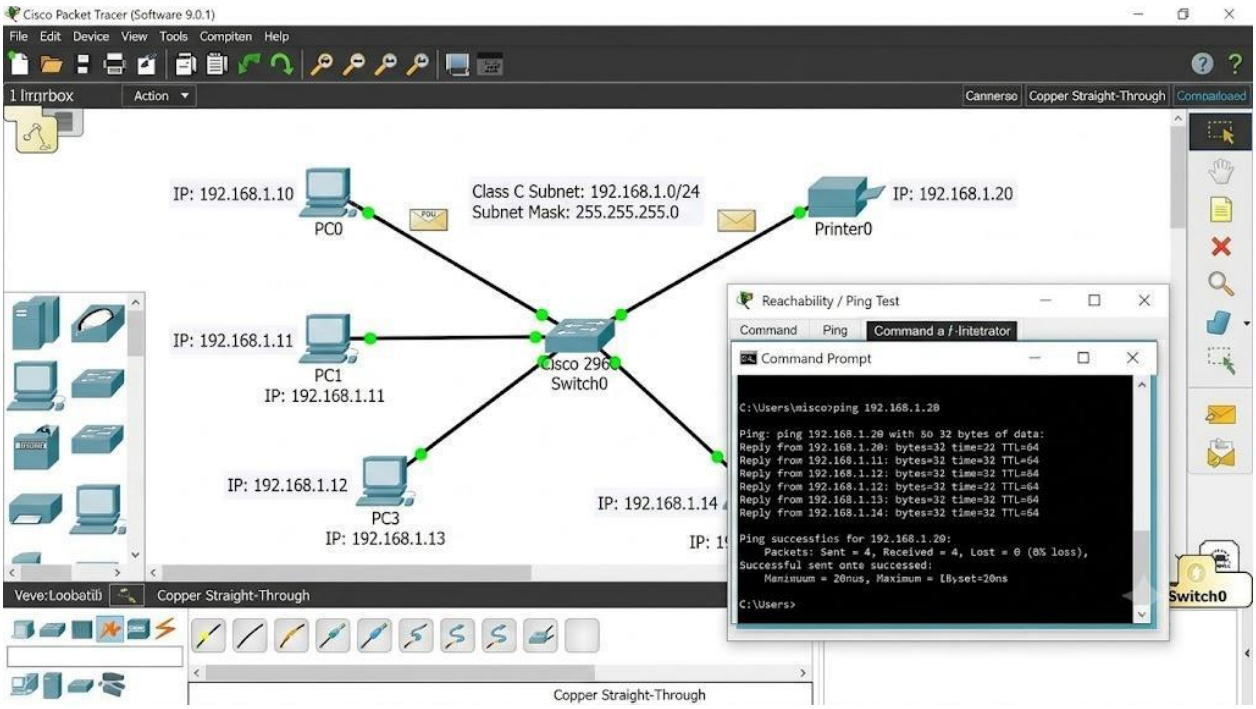
3. **Verify Specs:** Matched a pair of identical sticks with equal capacity, frequency, and latency values.

4. **Reseat Modules:** Firmly inserted the matched pair into the recommended dual-channel slots (slots 2 & 4) until retention clips clicked shut.

5. **BIOS Verification:** Booted into UEFI/BIOS to confirm both sticks were recognized at target speeds in dual-channel mode.

6. **Confirm Normal Boot:** Saved changes and verified the workstation passed POST cleanly straight into the operating system.

Task 3: LAN Topology Build



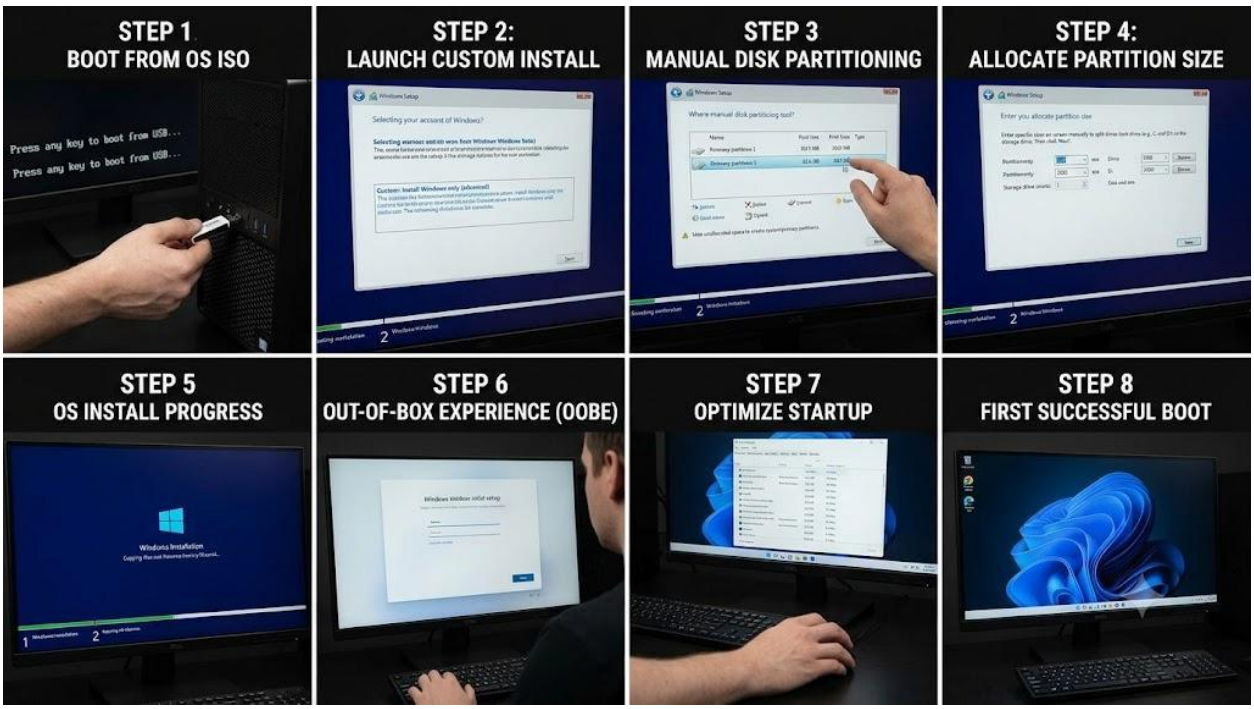
Topology Setup: Created a star LAN topology in Cisco Packet Tracer containing 5 desktop computers (PC0 to PC4) and 1 network printer (Printer0) connected to a central Cisco 2960 switch using copper straight-through cables.

Subnet Addressing: Assigned static Class C IP addresses to all devices on the 192.168.1.0/24 subnet (subnet mask 255.255.255.0):

- PCs: 192.168.1.10 through 192.168.1.14
- Network Printer: 192.168.1.20

Connectivity Verification: Demonstrated full reachability through ping tests (e.g., from PC0 to the printer) with a 0% packet loss rate. Let me know if you need to adjust or extend any configuration.

Task 4: Clean OS Installation



1. **Installation Prep & Setup (Steps 1–2):** Booted the workstation from a USB drive containing the Windows ISO and initiated the setup, selecting "Custom: Install Windows only" to manually manage the drive.

2. **Drive Partitioning (Steps 3–4):** Manually configured the disk by deleting existing volumes and partitioning the unallocated space to split the system and data drives, designating sizes before installing the OS.

3. **Completion & Personalization (Steps 5–6):** Monitored the automated OS installation, then configured localization, network, user accounts, and security preferences during the post-installation Out-of-Box Experience (OOBE).

4. **Optimization & Verification (Steps 7–8):** Opened Task Manager to disable unnecessary startup apps for faster boot times, then verified a quick, error-free boot to the newly configured desktop.

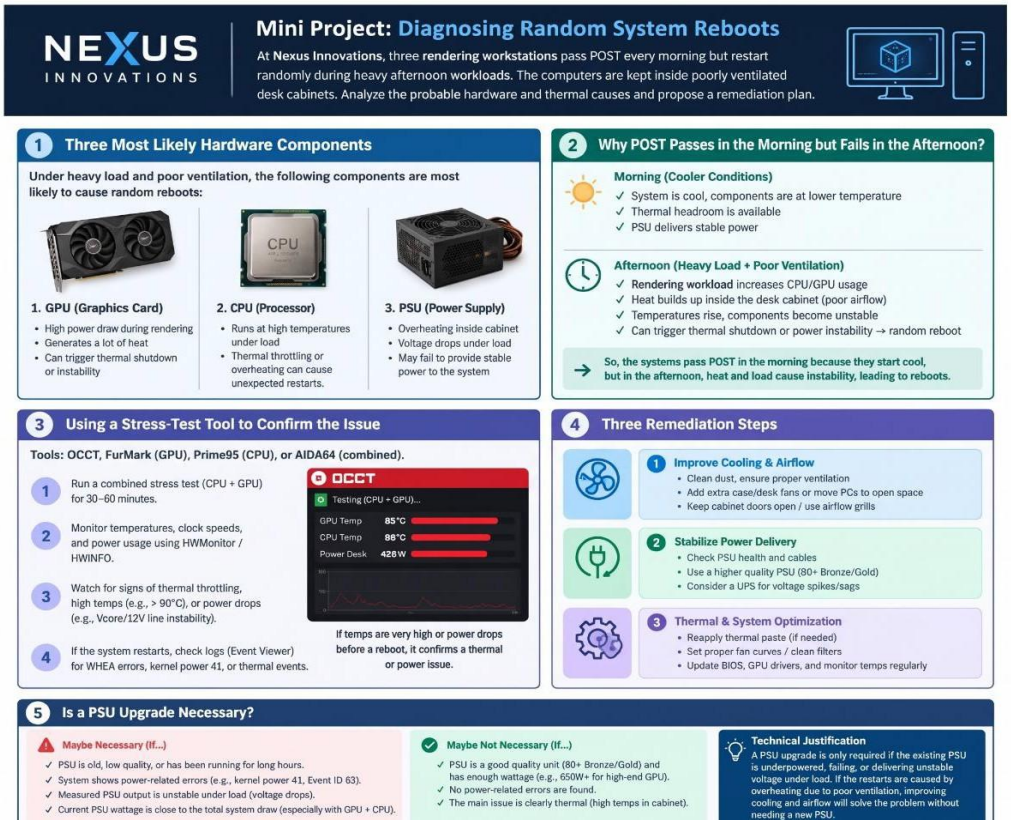
Section C

Mini Capstone Project

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

Objective:

At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.



1. Three Hardware Components Most Likely to Cause Random Reboots

CPU: Heavy rendering creates heat. Poor ventilation can cause the CPU to overheat and restart the computer.

Power Supply Unit (PSU): The PSU may not provide enough stable power during heavy workloads, causing sudden reboots.

RAM: Faulty or unstable RAM can cause system crashes and unexpected restarts during demanding tasks.

2. Why The Systems Pass Post in the Morning but Fail in The Afternoon.

- The systems pass **POST** in the morning because the components are cool and can start normally. In the afternoon, heavy rendering generates more heat, and the poorly ventilated desk cabinets trap hot air. The **CPU** or **GPU** may overheat, or the **power supply** may struggle under **heavy load**, causing the computers to restart randomly.

3. **Describe how you would use a stress-test tool to confirm a thermal or power issue.**

Monitor temperatures: Use a tool like HWMonitor to check CPU and GPU temperatures.

Run a stress test: Use a tool like OCCT to put the CPU and GPU under heavy load.

Check power stability: Monitor for sudden reboots, crashes, or power-related errors during the test.

Check results: If temperatures become too high before a reboot, overheating is likely. If the system restarts under load without excessive temperatures, the PSU or power delivery may be the cause.

Take action: Improve cabinet ventilation, clean the fans, and test the PSU if the problem continues.

4. **Propose three specific steps to remediate the frequent system restarts.**

Improve ventilation: Move the computers out of poorly ventilated cabinets, clean the fans, and ensure proper airflow to prevent overheating.

Check the power supply: Test the PSU and replace it if it cannot provide stable power under heavy workloads. Use a UPS to protect against power fluctuations.

Test hardware: Run CPU/GPU stress tests and check the RAM for errors. Replace any faulty components and monitor the systems for further restarts.

5. **Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.**

A PSU upgrade may be necessary if the current power supply cannot provide enough stable power during heavy rendering workloads, causing random restarts. Before upgrading, test the PSU and check its wattage, condition, and power connections. If the PSU is working properly and temperatures are too high, improving cooling may solve the problem without an upgrade.

Section D

AI-Augmented Learning

STEP 1 · BUILD WITH AI

Exact prompt used:

Create an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 computers and 1 printer. Specify the network subnet, subnet mask, usable host range, and broadcast address. Assign a unique IP address to each computer and the printer. Also identify IP addresses reserved for future expansion. Keep the answer simple and easy to understand.

IP Addressing Plan:

Network detail	Value
Network subnet	192.168.1.0/24
Subnet mask	255.255.255.0
Usable host range	192.168.1.1 – 192.168.1.254
Broadcast address	192.168.1.255
Default gateway	192.168.1.1

Device IP Allocation:

Device	**IP address**
System 1	192.168.1.10
System 2	192.168.1.11
System 3	192.168.1.12
System 4	192.168.1.13
System 5	192.168.1.14
Office Printer	192.168.1.20

Future Expansion:

Reserve 192.168.1.21 to 192.168.1.50 for future computers, printers, or other office devices.

Conclusion: The /24 subnet provides 254 usable host addresses, which is enough for the current six devices and future office expansion.

STEP 2 · TEST & DEBUG (WITHOUT AI)

Exact prompt given to the AI

Create an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 computers and 1 printer. Specify the network subnet, subnet mask, usable host range, and broadcast address. Assign a unique IP address to each computer and the printer. Also identify IP addresses reserved for future expansion. Keep the answer simple and easy to understand.

I's original plan and my corrected version

After checking the network address, host range, broadcast address, and reserved addresses, I found no subnetting errors in the original plan. The corrected version remains the same.

Detail	Verified plan
Network subnet	192.168.1.0/24
Subnet mask	255.255.255.0
Usable host range	192.168.1.1 to 192.168.1.254
Broadcast address	192.168.1.255
Default gateway	192.168.1.1
Total usable addresses	254

Device	Assigned IP
System 1	192.168.1.10
System 2	192.168.1.11
System 3	192.168.1.12
System 4	192.168.1.13
System 5	192.168.1.14
Printer	192.168.1.20

Reserved for future expansion: 192.168.1.21 to 192.168.1.50.

Network address: 192.168.1.0, not assignable to a device.

Broadcast address: 192.168.1.255, not assignable to a device.

Gateway: 192.168.1.1, excluded from device assignments.

My verification notes

I checked the subnet mask, usable host range, network address, and broadcast address. The /24 subnet provides 254 usable IP addresses, which is sufficient for the office network. All six devices have unique addresses, and the future expansion range does not overlap with their assigned addresses.
Therefore, no corrections were necessary because the AI's original plan was already correct.